

Does a VAR Beat a Univariate AR at Forecasting Inflation?

Evidence from the Euro Area and the United States

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The question

Intuition. More information should sharpen an inflation forecast: unemployment and interest rates carry signal (Phillips curve, monetary policy). The **vector autoregression (VAR)** is the canonical tool to use it [Sims 1980].

But. Simple **univariate** models are famously hard to beat for inflation [Atkeson & Ohanian 2001].

Research question

Does a VAR in *inflation, unemployment and a short-term interest rate* forecast inflation more accurately than a univariate AR — and does the answer depend on the economy?

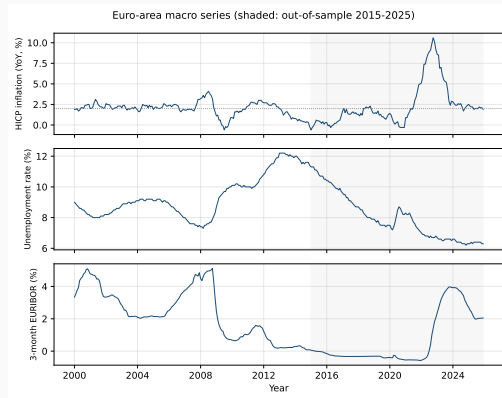
A clean test of whether cross-series information *pays for the parameter cost* of a VAR. We answer it on **two** economies to check robustness across monetary regimes.

Three monthly series, 2000–2025

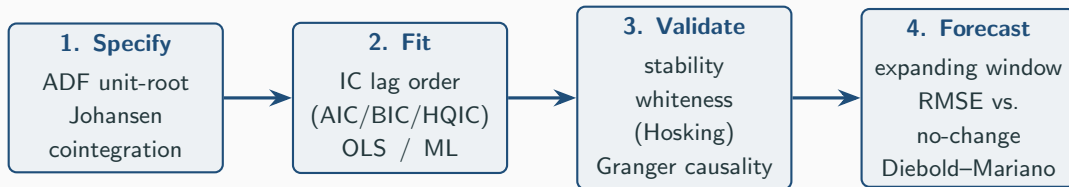
- **Euro area** (ECB): HICP inflation, unemployment rate, 3-month EURIBOR
- **United States** (FRED): CPI YoY inflation, UNRATE, federal funds rate

Leakage-free out-of-sample design

- Train 2000–2014 ($T = 180$ months)
- Forecast 2015–2025, **expanding window**, models refit at every origin
- All series modelled in **first differences**



Methodology: one pipeline, run on both economies



- Software: Python statsmodels; identical code, only the data-loading step changes.
- Benchmark: a **no-change** random walk — a demanding baseline for inflation.
- Diebold–Mariano uses a Newey–West (HAC) variance; we test it at the **one-step** horizon.

The AR is nested in the VAR at matched lag

$$X_t = c + \Phi_1 X_{t-1} + \cdots + \Phi_p X_{t-p} + Z_t$$

Setting the cross-variable coefficients to zero gives the AR(p).

The parameter cost is quadratic

- VAR(p) has $p d^2$ slope coefficients, $d = 3$
- Seasonality needs a lag of 12:
 - AR: 12 coefficients
 - VAR: $12 \times 3^2 > 100$

The bet we are testing

Does the extra cross-series information buy more accuracy than the estimation noise its parameters cost?

Level forecasts are recovered by accumulating the forecast *changes*; accuracy is judged out of sample, never on in-sample fit.

In-sample lag selection

Pick the lag order by information criteria

- Each criterion trades fit against a penalty $\propto p d^2$
- BIC, HQIC are consistent for the true order; AIC tends to **overfit**
- We prefer **parsimony** (HQIC)

Euro-area VAR: AIC, FPE, HQIC point to **VAR(2)**, BIC to 1.

Univariate AR (differenced inflation):
AIC, HQIC $\rightarrow$ **AR(12)** (seasonal), BIC $\rightarrow$ 2.

Euro-area VAR order selection (* = minimum)

p	AIC	BIC	FPE (10^{-6})	HQIC
0	-11.35	-11.29	11.82	-11.32
1	-12.16	-11.94*	5.27	-12.07
2	-12.26*	-11.88	4.74*	-12.11*
3	-12.20	-11.66	5.03	-11.98
4	-12.20	-11.50	5.03	-11.92
5	-12.13	-11.25	5.43	-11.77
6	-12.09	-11.05	5.65	-11.66

Lower is better. AIC/FPE/HQIC minimise at $p = 2$, BIC at $p = 1$.

In-sample tests: are the modelling choices justified?

Test	Question	Euro-area answer
ADF unit root	Stationary in levels?	No in levels, yes in differences $\Rightarrow$ model in first differences
Johansen trace	Cointegration?	Borderline (30.76 vs. 29.80) $\Rightarrow$ keep VAR-in-differences
Ljung–Box / Hosking	Residuals white?	AR(12): yes; VAR(2): no (lag-12 spike)
Granger causality	Do unemployment & rate help inflation?	$p = 0.456 \Rightarrow$ no added information

The tests are the *justification* for the modelling choices, and they validate the out-of-sample comparison. The insignificant Granger test already hints at the final answer.

Out-of-sample evaluation

Expanding window (no look-ahead)

- At each origin t : refit on all data up to t , forecast h months ahead
- Fit on differences, then recover the level by summing the forecast changes:

$$\hat{\pi}_{t+h} = \pi_t + \sum_{i=1}^h \widehat{\Delta\pi}_{t+i}$$

- Accuracy: **RMSE** and MAE of the level forecast
- Benchmark: **no-change** random walk
$$\hat{\pi}_{t+h} = \pi_t$$

Diebold–Mariano test

Tests equal accuracy of two forecasts.
With loss differential $d_t = e_{1,t}^2 - e_{2,t}^2$,

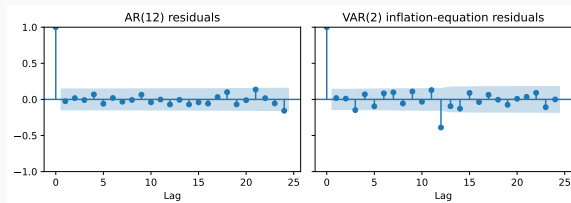
$$DM = \frac{\bar{d}}{\sqrt{\hat{\omega}^2/n}} \xrightarrow{d} \mathcal{N}(0, 1),$$

where $\hat{\omega}^2$ is a Newey–West (HAC) variance. We test it at the **one-step** horizon.

Euro Area

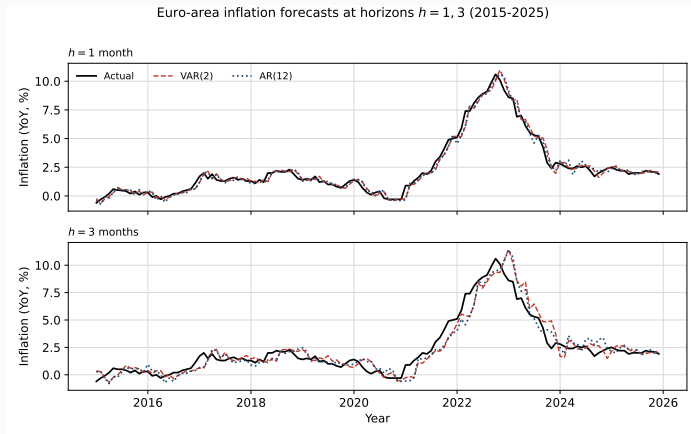
Euro area: specification & diagnostics

- **ADF:** levels non-stationary, differences stationary $\Rightarrow$ model in differences
- **Johansen:** borderline (trace 30.76 vs. 29.80) $\Rightarrow$ keep VAR-in-differences
- **Lag choice:** seasonal **AR(12)**; parsimonious **VAR(2)**
- **Granger** (unemp, rate $\rightarrow$ inflation): $p = 0.456$ — insignificant
- VAR(2) residuals not white: spike left at lag 12



Residual ACF: AR(12) clean (left), VAR(2) keeps a lag-12 spike (right).

Euro area: the forecasts are nearly identical



VAR(2) and AR(12) track inflation almost identically at $h = 1, 3$ and both lag the 2021–22 surge.

Euro area: accuracy and verdict

Out-of-sample RMSE (best in **bold**)

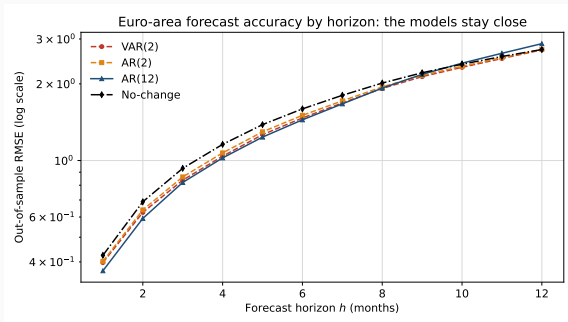
h	VAR(2)	AR(12)	No-ch.
1	0.399	0.369	0.425
3	0.842	0.820	0.932
6	1.479	1.445	1.595
12	2.728	2.876	2.728

Diebold–Mariano ($h = 1$)

- VAR vs. AR(12): $p = 0.16$ (tie)
- AR(12) vs. no-change: $p = 0.067$

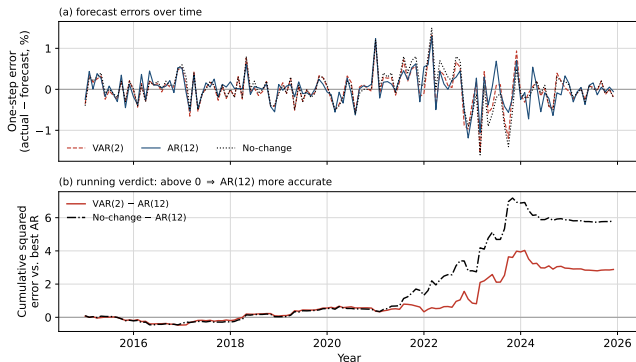
Euro-area verdict

No significant evidence the VAR beats the AR. The seasonal **AR(12)** is best at $h = 1-6$; gains over no-change are only marginal.



Euro area: who wins over time?

Euro-area one-step forecast errors and running verdict (2015-2025)



Because the forecasts overlap so closely, we look at the **errors** directly.

Panel (b) adds up the squared-error gap of AR(12) over the VAR and the no-change benchmark. It stays near zero until 2021 and then grows only a little during the surge.

A **small, late** edge for AR(12), and no systematic winner.

United States

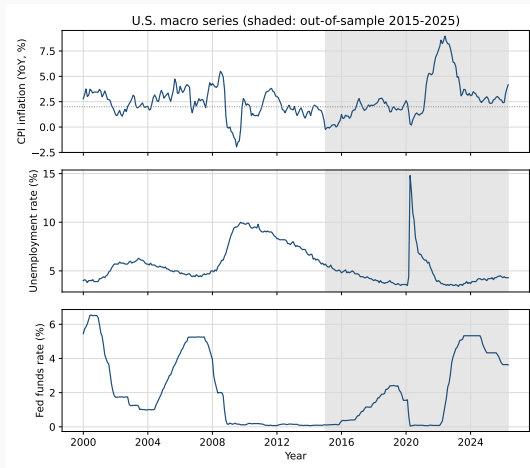
United States: does the verdict travel?

Same three variables, same pipeline, same 2000–2014 / 2015–2025 split.

- **ADF:** model in **first differences** again, as in the euro area
- **Johansen:** **rank 0** ($25.70 < 29.80$) $\Rightarrow$ *no* cointegration (cleaner than the euro area)
- Best univariate model **AR(24)**, compared against the VAR

Keep an eye on 2020

The sharp COVID unemployment spike is the feature that will matter for the VAR.

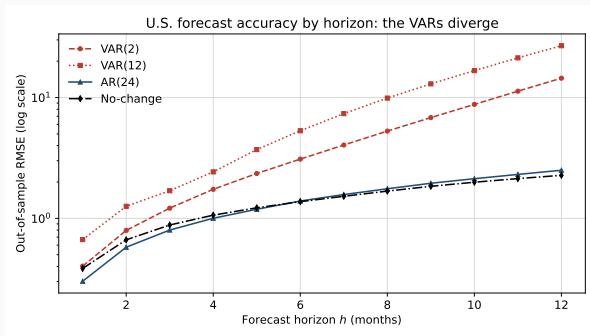


United States: same answer, stronger

Out-of-sample RMSE

h	VAR	AR(24)	No-ch.
1	0.402	0.302	0.385
3	1.215	0.800	0.881
6	3.093	1.392	1.375
12	14.5	2.502	2.270

- The univariate **AR(24)** is the most accurate
- The VAR **blows up** at long horizons: the 2020 shock feeds through the differenced VAR (any lag length)



Same conclusion as the euro area, only stronger: more variables do not help, and the VAR can badly hurt the forecast.

Findings, limitations, extensions

Results of our work

Euro area

- VAR does *not* beat the AR at any horizon
- Best model = seasonal **AR(12)** (short/mid horizons)
- Mechanism: seasonality is cheap univariately, unaffordable in a VAR

United States

- Same verdict, stronger: the VAR does not help
- At long horizons the VAR **blows up**

The verdict travels

Across two economies with different monetary regimes, adding unemployment and an interest rate to a univariate model of inflation **does not improve, and can badly degrade, the forecast.**

Takeaway

More variables $\neq$ better forecasts. A parsimonious univariate model is a strong, hard-to-beat benchmark for inflation.

Limitations

- **One split, revised data.** A single training/evaluation window and final-vintage (not real-time) data — a mild, equal advantage to all models.
- **Significance scoped to $h = 1$.** Diebold–Mariano is applied only one-step; longer horizons are read descriptively.
- **The COVID window.** 2020 shock + 2021–23 surge inflate error levels and strain the constant-parameter assumption; single-window results should be read with this in mind.
- **Borderline cointegration (EA).** A VECM might behave differently at long horizons.
- **Linear, fixed coefficients.** Every model here missed the surge — the architecture, not the univariate-vs-multivariate choice, may be the binding constraint.

Possible extensions

- **Cointegration.** A VECM for the borderline euro-area system; compare long-horizon behaviour.
- **Sharper tests.** Clark–West for the *nested* comparison; multi-horizon Diebold–Mariano.
- **Relax the architecture.** Time-varying-parameter and regime-switching state-space models (Kalman filter) for the 2021–23 regime shift.
- **Nonlinear / ML methods** — with the caveat that evidence they beat simple univariate benchmarks for inflation remains mixed [Faust & Wright 2013].
 - **Real-time data** vintages, to remove the revised-data advantage.

The binding question is whether a *richer architecture* — not simply more variables — can capture regime-dependent inflation dynamics.

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Thank you!

Questions?

Code: `github.com/ZhukovAndrii/MTSA\_project`